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Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling

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Research Goal

■ Few-Shot Multi-Objective Optimization

- $\operatorname{argmin}_{x \in \mathcal{X}} F(x) = \operatorname{argmin}_{x \in \mathcal{X}} (f_1(x), \dots, f_M(x))$
- The evaluation budget for the optimization is small

■ How?

Multi-Objective Multi-Task Optimization[1]

Previous Work 1

- $\operatorname{argmin}_{x_k \in \mathcal{X}_k} F_k(x_k), m \in [K]$

Assuming K multi-objective problems share certain similarities, we use their **Pareto-optimal solution distributions and surrogate models** as transferable knowledge to accelerate convergence.

Parametric Multi-Task Optimization[2]

Previous Work 2

- $\operatorname{argmin}_{x \in \mathcal{X}} f(x, \theta), \forall \theta \in \Theta$

We assume access to **an external task space Θ** and investigate how leveraging **task parameters θ as side information** can enhance knowledge transfer and accelerate optimization convergence.

Parametric Multi-Objective Multi-Task Optimization[3]

This Work

- $\operatorname{argmin}_{x \in \mathcal{X}} F(x, \theta), \forall \theta \in \Theta$

We assume access to **an external task space Θ** and investigate how leveraging **task parameters θ as side information** can accelerate optimization convergence and **return any preferred solution observing preference vector λ in any task θ** .

[1]. Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “Bayesian Forward-Inverse Transfer for Multiobjective Optimization” International Conference on Parallel Problem Solving from Nature 2024.

[2]. Tingyang Wei, Jiao Liu, Abhishek Gupta, Puay Siew Tan, Yew-Soon Ong, “ (Θ_l, Θ_u) -Parametric Multi-Task Optimization: Joint Search in Solution and Infinite Task Spaces” IEEE Transactions on Evolutionary Computation (2025).

[3]. Tingyang Wei, Jiao Liu, Abhishek Gupta, Chin Chun Ooi, Puay Siew Tan, Yew-Soon Ong, “Amortized Multi-Objective Optimization Across Tasks with Generative Solution Modeling”, IJCAI 2026.

Why Parametric Multi-Objective Multi-Task Optimization?

$$\blacksquare \text{ } \argmin_{x \in \mathcal{X}} F(x, \theta) := (f_1(x, \theta), \dots, f_M(x, \theta)), \forall \theta \in \Theta$$

Opportunities:

- Explicit task parameters θ provide a structured way to **quantify task similarity**, contributing to **faster multi-objective optimization convergence** via efficient knowledge transfer.

Challenges:

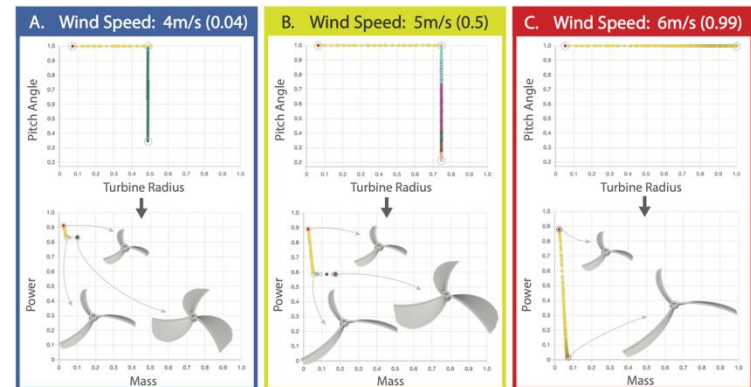
- The task parameters θ lie in a continuous space, meaning we have **infinitely many** multi-objective tasks to solve, and each multi-objective task require **potentially infinitely many solutions** to satisfy **arbitrary preference articulation** on Pareto Front for **any task**.

Application Example:

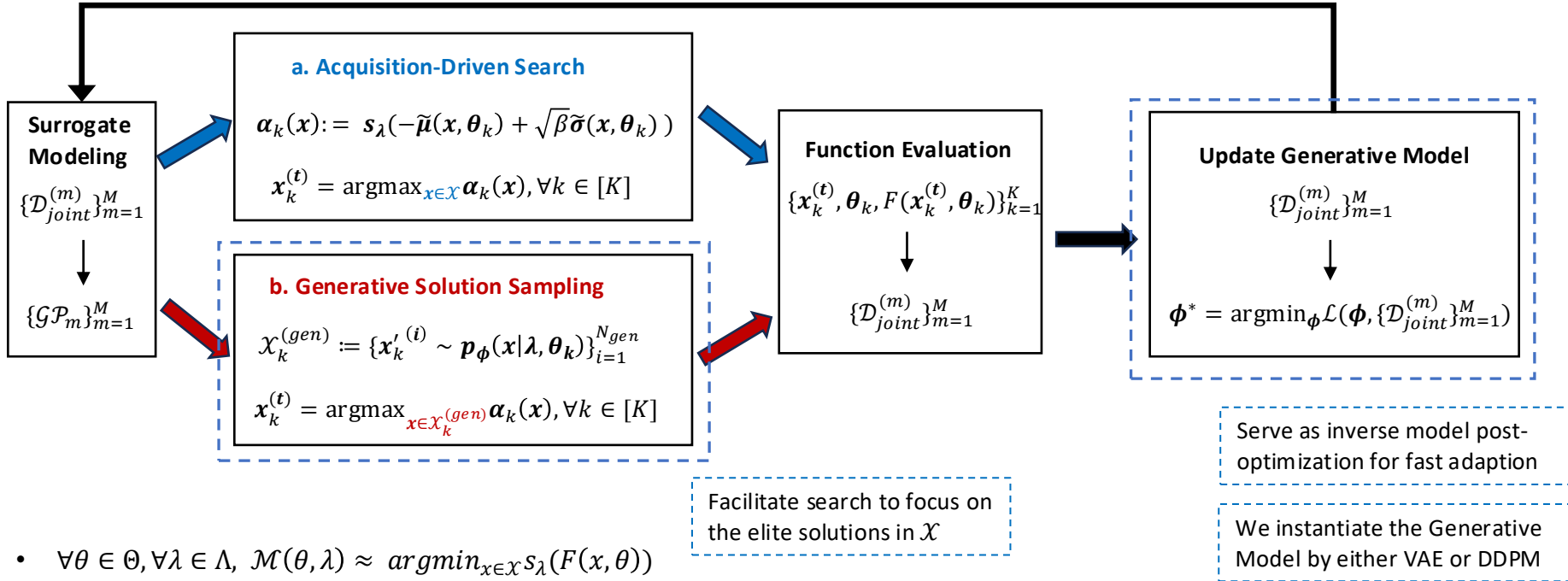
- In turbine design, θ can represent different **wind speeds**. Optimizing manufactured components x for diverse θ allows the parametric turbine to **adapt to unexpected environments**.

$$\forall \theta \in \Theta, \forall \lambda \in \Lambda$$
$$\mathcal{M}(\theta, \lambda) \approx \argmin_{x \in \mathcal{X}} s_{\lambda}(F(x, \theta))$$

λ means preference vector to trade-off the objectives



Methodology



- $\forall \theta \in \Theta, \forall \lambda \in \Lambda, \mathcal{M}(\theta, \lambda) \approx \operatorname{argmin}_{x \in \mathcal{X}} s_\lambda(F(x, \theta))$

- Surrogate Models:**

- $\mathcal{GP}: \mathcal{X} \times \Theta \rightarrow \mathbb{R}$
- (M models for M objectives)

Facilitate cross-task exploration in $\mathcal{X} \times \Theta$

- Cross-Task Sampling:**

- Forward Model contains cross-task information:
- $\mathcal{GP}(\mu(\cdot, \theta), \sigma^2(\cdot, \theta))$

- Conditional Generative Model (Inverse Model):**

- $\mathcal{M}_\phi: \Theta \times \Lambda \rightarrow \mathcal{X}$
- (Identify the preferred solution by λ for task θ)

- Alternating Optimization Loop:**

- Per iteration, mode **a.** and **b.** alternates in a round robin way

Results

- $\operatorname{argmin}_{x \in \mathcal{X}} F(x, \theta) := (f_1(x, \theta), \dots, f_M(x, \theta)), \forall \theta \in \Theta$
- Good Optimization Results
 - Sample K multi-objective tasks
 - Check out the multi-objective convergence
 - $F(\cdot, \theta_k), k \in [K]$
 - Theoretical Analysis of potentially faster convergence

Theorem 2 (Tightened Regret Bounds for PMT-MOBO). Under Assumption 1 and with the task-aware kernel that satisfies: $\tilde{\kappa}_m((\mathbf{x}, \theta_k), (\mathbf{x}', \theta_k)) = \kappa_m(\mathbf{x}, \mathbf{x}'), \forall k \in [K]$, let $\tilde{\gamma}_{T,k} = \max_m \tilde{\gamma}_{T,m,k}$ be the largest MIG of PMT-MOBO across M objective functions over T scalarized evaluations for the k -th task, it holds that

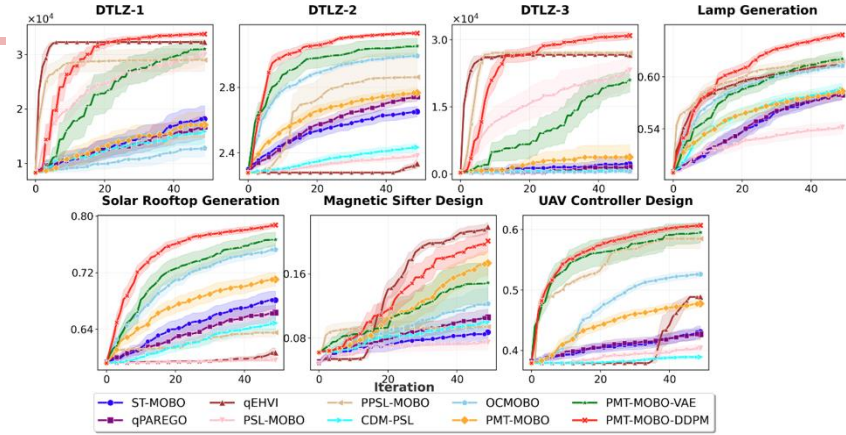
$$\tilde{\gamma}_{T,m,k} \leq \gamma_{T,m,k}, \forall m \in [M], \forall k \in [K]. \quad (13)$$

Accordingly, it also holds that $\tilde{\gamma}_{T,k} \leq \gamma_{T,k}, \forall k \in [K]$. Consider that both regret bounds from Theorem 1 still hold with $\gamma_{T,k}$ replaced by $\tilde{\gamma}_{T,k}$, consequently, it holds that,

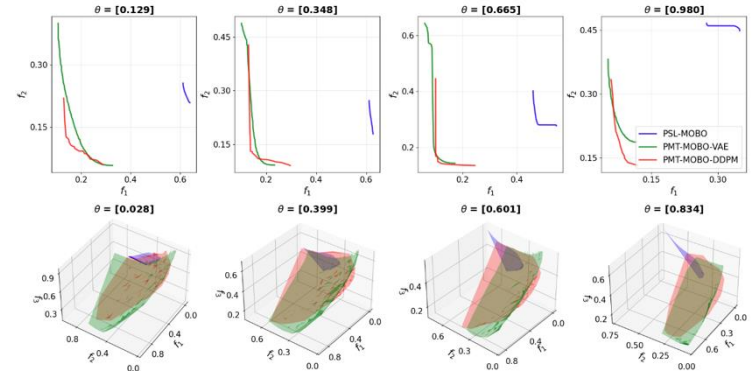
$$\mathbb{E}[\mathcal{R}_B^{\text{PMT}}(T)] \leq \mathbb{E}[\mathcal{R}_B(T)], \quad (14)$$

i.e. PMT-MOBO achieves a tighter upper bound on Bayes regret than standard ST-MOBO.

- $\forall \theta \in \Theta, \forall \lambda \in \Lambda, \mathcal{M}(\theta, \lambda) \approx \operatorname{argmin}_{x \in \mathcal{X}} s_\lambda(F(x, \theta))$
- Good Inverse Model Predictive Results
 - Sample multiple (unseen) multi-objective tasks θ
 - Sample diverse preference vectors $\lambda \in \Lambda$
 - Check out the solution quality of the model output
 - $F(x', \theta), x' = \mathcal{M}_\phi(\theta, \lambda),$



Our PMT-MOBO-VAE/DDPM exhibit better hypervolume results



Our PMT-MOBO-VAE/DDPM generate better Pareto-optimal solutions

Detailed Comparisons

Method	DTLZ-1	DTLZ-2	DTLZ-3	LAMP	SOLAR	UAV
PSL-MOBO	2.82e+04 (1.78e+03) ^{≈-}	1.90e+00 (1.30e-01) ⁻⁻	2.39e+04 (3.22e+03) ⁺⁻	5.01e-01 (7.32e-02) ⁻⁻	3.54e-01 (4.06e-02) ⁻⁻	4.12e-01 (7.30e-03) ⁻⁻
CDM-PSL	1.92e+04 (3.85e+03) ⁻⁻	2.52e+00 (8.49e-02) ⁻⁻	1.83e+03 (1.31e+03) ⁻⁻	6.19e-01 (4.45e-02) ⁻⁻	6.54e-01 (3.11e-02) ⁻⁻	4.26e-01 (3.09e-02) ⁻⁻
PPSL-MOBO	2.24e+04 (1.10e+02) ⁻⁻	2.98e+00 (3.56e-02) ⁻⁻	2.67e+04 (6.08e+02) ⁺⁻	6.12e-01 (1.05e-01) ^{≈-}	4.86e-01 (1.11e-01) ⁻⁻	5.24e-01 (1.71e-01) ⁻⁻
PMT-MOBO-VAE	2.88e+04 (2.57e+03)	3.10e+00 (2.99e-02)	1.78e+04 (7.28e+03)	6.31e-01 (7.33e-02)	7.93e-01 (1.56e-02)	6.08e-01 (1.88e-02)
PMT-MOBO-DDPM	3.47e+04 (4.85e+02)	3.13e+00 (1.41e-02)	3.20e+04 (1.98e+03)	6.70e-01 (5.82e-02)	7.65e-01 (1.30e-02)	6.05e-01 (2.44e-02)

Table 1: Inverse Model Generalization Performance on Unseen EMOP Tasks. Statistical significance is assessed using the Wilcoxon signed-rank test at the 95% confidence level. Superscripts ^{s1s2} indicate that the competitor is significantly worse (−), significantly better (+), or statistically similar (≈) relative to **PMT-MOBO-VAE** (s_1) and **PMT-MOBO-DDPM** (s_2).

Method	DTLZ-1	DTLZ-2	DTLZ-3	LAMP	SOLAR	MAGNETIC	UAV
ST-MOBO	1.82e+04 (4.80e+03) ⁻⁻	2.65e+00 (6.30e-02) ⁻⁻	2.24e+03 (3.03e+03) ⁻⁻	5.80e-01 (1.20e-02) ⁻⁻	6.81e-01 (2.51e-02) ⁻⁻	8.72e-02 (2.15e-02) ⁻⁻	4.31e-01 (1.99e-02) ⁻⁻
qPAREGO	1.68e+04 (4.03e+03) ⁻⁻	2.74e+00 (5.30e-02) ⁻⁻	1.42e+03 (1.48e+03) ⁻⁻	5.79e-01 (1.30e-02) ⁻⁻	6.64e-01 (2.21e-02) ⁻⁻	1.06e-01 (2.05e-02) ⁻⁻	4.27e-01 (1.82e-02) ⁻⁻
qEHVI	3.23e+04 (5.06e+01) ^{≈-}	2.33e+00 (6.20e-02) ⁻⁻	2.66e+04 (1.85e-02) ⁺⁻	6.14e-01 (3.00e-03) ⁻⁻	6.07e-01 (1.29e-02) ⁻⁻	2.18e-01 (1.15e-02) ⁺⁺	4.89e-01 (2.60e-03) ⁻⁻
PSL-MOBO	3.00e+04 (3.52e+03) ^{≈-}	2.38e+00 (4.90e-02) ⁻⁻	2.31e+04 (6.44e+03) ^{≈-}	5.42e-01 (1.40e-02) ⁻⁻	5.96e-01 (3.60e-03) ⁻⁻	7.42e-02 (1.79e-02) ⁻⁻	4.03e-01 (1.46e-02) ⁻⁻
CDM-PSL	1.59e+04 (4.15e+03) ⁻⁻	2.43e+00 (5.04e-02) ⁻⁻	7.76e+02 (9.41e+02) ⁻⁻	5.85e-01 (1.39e-02) ⁻⁻	6.49e-01 (1.97e-02) ⁻⁻	1.00e-01 (1.82e-02) ⁻⁻	3.89e-01 (5.70e-03) ⁻⁻
PPSL-MOBO	3.20e+04 (1.72e+03) ^{≈-}	2.86e+00 (3.00e-01) ^{≈-}	2.71e+04 (7.89e+02) ⁺⁻	6.09e-01 (1.61e-02) ⁻⁻	6.35e-01 (1.15e-02) ⁻⁻	9.41e-02 (1.69e-02) ⁻⁻	5.85e-01 (3.19e-02) ^{≈-}
OCMOBO	1.28e+04 (3.04e+03) ⁻⁻	2.99e+00 (2.12e-02) ^{≈-}	6.20e+02 (6.29e+02) ⁻⁻	6.13e-01 (6.60e-03) ⁻⁻	7.53e-01 (1.36e-02) ⁻⁻	1.23e-01 (3.95e-02) ⁻⁻	5.26e-01 (1.09e-02) ⁻⁻
PMT-MOBO	1.71e+04 (4.70e+03)	2.77e+00 (5.50e-02)	3.78e+03 (5.40e+03)	5.83e-01 (1.22e-02)	7.10e-01 (1.90e-02)	1.73e-01 (2.99e-02) ^{≈-}	4.78e-01 (1.27e-02)
PMT-MOBO-VAE	3.10e+04 (3.41e+03)	3.05e+00 (9.58e-02)	2.10e+04 (5.33e+03)	6.21e-01 (1.73e-02)	7.67e-01 (2.01e-02)	1.49e-01 (5.07e-02)	5.95e-01 (3.33e-02)
PMT-MOBO-DDPM	3.37e+04 (8.35e+02)	3.13e+00 (1.62e-02)	3.09e+04 (1.67e+03)	6.48e-01 (5.10e-03)	7.87e-01 (7.80e-03)	2.01e-01 (2.50e-02)	6.07e-01 (7.50e-03)

Table 5: Average Hypervolume results (mean $\pm$ std) across synthetic and real-world benchmarks. Statistical significance is assessed using the Wilcoxon signed-rank test at the 95% confidence level. Superscripts ^{s1s2} indicate that the competitor is significantly worse (−), significantly better (+), or statistically similar (≈) relative to **PMT-MOBO-VAE** (s_1) and **PMT-MOBO-DDPM** (s_2).

Conclusion

- We aim to provide a framework extending multitask optimization to infinite multi-task optimization.
- In TEVC2025, we present the parametric multitask optimizer for single objective optimization.
- In IJCAI2026, we present an alternation-based strategy to build a parametric multiobjective multitask optimizer based on generative models.

Thank you for your attention!

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